

30 Meaningful Showcase Questions & Answers Guide

Northgate Energy / Refinery AI Knowledge Copilot Reference Guide

Use this PDF guide during your hackathon presentation or project demonstration. You can ask any of these 30 questions directly to the **Expert RAG Copilot** (<http://localhost:5173>) and get instant, cited answers!

■ ■ Section 1: Refinery Working & Basics (Q1 – Q5)

Q1: "How does an oil refinery work in simple terms?"

- **Expected AI Answer:** Explains that raw crude oil is boiled and separated into different fuels (LPG cooking gas, petrol, jet fuel, diesel, fuel oil, asphalt) across 7 main plant sections.
- **Source Citation:** 01_refinery_working_and_parts_guide.txt

Q2: "What is the refining capacity and history of Northgate Energy?"

- **Expected AI Answer:** Founded in 1988 (Phase 1: 5.0 MMTPA), expanded in 1996 to 15.2 MMTPA, added BS-VI clean fuel unit in 2021, and integrated AI Knowledge Brain in 2024.
- **Source Citation:** 01_refinery_working_and_parts_guide.txt

Q3: "What are the 7 main operational sections of the refinery?"

- **Expected AI Answer:** 1. Tank Farm Storage, 2. Desalting Unit, 3. Primary Distillation (Tower T-101), 4. Vacuum Distillation (Column T-102), 5. Hydrocracker Unit, 6. Utilities & Boilers, 7. Pipe Racks & Flare Stack.
- **Source Citation:** 01_refinery_working_and_parts_guide.txt

Q4: "What does the Desalter unit DS-01 do and why is it important?"

- **Expected AI Answer:** Washes salt, sand, and water out of raw crude oil using electric fields to prevent pipe rust and corrosion.
- **Source Citation:** 01_refinery_working_and_parts_guide.txt

Q5: "What is the role of the Hydrocracker Reactor R-201?"

- **Expected AI Answer:** Takes cheap, heavy leftover oil and cracks (breaks) large molecules into high-value petrol and diesel under high heat, hydrogen gas, and pressure.
 - **Source Citation:** 01_refinery_working_and_parts_guide.txt
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■ ■ Section 2: Plant Geography & Map Locations (Q6 – Q10)

Q6: "can u give me a map"

- **Expected AI Answer:** Provides a visual breakdown of Zones A through F (Crude, Hydrocracker, Boiler House, Tank Farm, Pipe Racks, CCR) and directs you to the live blueprint map.
- **Source Citation:** 02_plant_map_and_zones_guide.txt

Q7: "Where is Feed Gas Compressor C-102 located?"

- **Expected AI Answer:** Zone B (Hydroprocessing Area) at Grid C-08 in Compressor House Bay 2.
- **Source Citation:** 02_plant_map_and_zones_guide.txt

Q8: "Where is High-Pressure Boiler B-201 located?"

- **Expected AI Answer:** Zone C (Utilities & Power Generation) at Grid S-22 in Boiler Building 1.
- **Source Citation:** 02_plant_map_and_zones_guide.txt

Q9: "Where are the crude oil storage tanks located?"

- **Expected AI Answer:** Zone D (Tank Farm) at Grid SW-02 to SW-06 (Tanks T-501 to T-503).
- **Source Citation:** 02_plant_map_and_zones_guide.txt

Q10: "Where is Safety Muster Point 4 located?"

- **Expected AI Answer:** Grid C-13 at Hydrocracker East Gate for Zone B assembly.
- **Source Citation:** 02_plant_map_and_zones_guide.txt

■ Section 3: Malfunctions, Repairs & Parts (Q11 – Q15)

Q11: "What is the breakdown and repair history of the refinery?"

- **Expected AI Answer:** Summarizes 5 major incidents (2018 Boiler B-201 soot, 2020 Tower T-101 tray flooding, 2022 Pump P-401B seal crack, 2023 Compressor C-102 valve wear, 2024 Exchanger EX-301 leak).
- **Source Citation:** 03_refinery_malfunctions_and_repairs_history.txt

Q12: "What issue was reported for Compressor C-102 in July 2023?"

- **Expected AI Answer:** Knocking sounds and pressure drop to 142 psi due to monsoon rain moisture entering lube oil. Valve plate CV-2210 was replaced.
- **Source Citation:** 03_refinery_malfunctions_and_repairs_history.txt

Q13: "What replacement part was used for Compressor C-102?"

- **Expected AI Answer:** Discharge valve plate Part# CV-2210.
- **Source Citation:** 03_refinery_malfunctions_and_repairs_history.txt

Q14: "What happened to Boiler Feed Pump P-401B in November 2022?"

- **Expected AI Answer:** Mechanical seal face cracked due to dry running. Standby Pump P-401A auto-started with zero downtime. Seal MS-8840 replaced.
- **Source Citation:** 03_refinery_malfunctions_and_repairs_history.txt

Q15: "What maintenance was done on Heat Exchanger EX-301 in Feb 2024?"

- **Expected AI Answer:** 3 leaking tube joints were plugged with brass tapered plugs and hydro-tested at 450 psi.
- **Source Citation:** 03_refinery_malfunctions_and_repairs_history.txt

■ Section 4: Lead Engineers & Supervision (Q16 – Q20)

Q16: "Which Lead Engineer is responsible for Boiler B-201?"

- **Expected AI Answer:** Rajesh Sharma (Lead Thermal & Utilities Engineer).
- **Source Citation:** 04_engineer_supervision_and_operating_log.txt

Q17: "Which Lead Engineer is responsible for Compressor C-102 and Pumps P-401B?"

- **Expected AI Answer:** Sarah Jenkins (Lead Mechanical Engineer - Rotating Equipment).
- **Source Citation:** 04_engineer_supervision_and_operating_log.txt

Q18: "Who is in charge of Process Quality and Relief Valve inspections?"

- **Expected AI Answer:** Maria Santos (Lead Process & Quality Inspection Engineer).
- **Source Citation:** 04_engineer_supervision_and_operating_log.txt

Q19: "Who authorizes Safety LOTO permits and emergency drills?"

- **Expected AI Answer:** Vikram Patel (Lead Safety & Emergency Response Officer).
- **Source Citation:** 04_engineer_supervision_and_operating_log.txt

Q20: "Who is the Chief Reliability & SCADA Director?"

- **Expected AI Answer:** Dr. Aris Thorne (SCADA Director & AI Systems Lead).
- **Source Citation:** 04_engineer_supervision_and_operating_log.txt

■ Section 5: Emergency Shutdown & Safety Protocols (Q21 – Q25)

Q21: "What are the emergency shutdown protocols (ESD Level 1 to 3)?"

- **Expected AI Answer:** ESD 1 = Full plant trip; ESD 2 = Unit isolation; ESD 3 = Depressurization via Flare Stack FL-01.
- **Source Citation:** 06_emergency_shutdown_and_safety_protocol.txt

Q22: "What happens during a toxic H2S gas leak?"

- **Expected AI Answer:** 10 ppm = Low alarm (amber beacon); 20 ppm = High alarm (SCBA mask mandatory, evacuate to Muster Point 4).
- **Source Citation:** 06_emergency_shutdown_and_safety_protocol.txt

Q23: "What are the rules for Lockout / Tagout (LOTO) safety permits?"

- **Expected AI Answer:** Disconnect electrical breaker at Substation 3, attach signed red padlock, and verify zero pressure/voltage before servicing.
- **Source Citation:** 06_emergency_shutdown_and_safety_protocol.txt

Q24: "When is the next inspection due for Pressure Relief Valve PRV-88?"

- **Expected AI Answer:** Due 12-Mar-2024 per OISD-116 standards.
- **Source Citation:** 05_master_plant_logbook_2023_2024.txt

Q25: "What is the emergency extension for the Fire Station?"

- **Expected AI Answer:** Extension 2222 (ECC Main Control: Ext 3333).
- **Source Citation:** 06_emergency_shutdown_and_safety_protocol.txt

■ Section 6: Product Quality, Environment & Logistics (Q26 – Q30)

Q26: "What is the required Octane rating for BS-VI petrol?"

- **Expected AI Answer:** Minimum 95.0 RON with maximum 10 ppm sulfur.
- **Source Citation:** 07_quality_assurance_and_fuel_testing_standards.txt

Q27: "What is the Cetane index requirement for High-Speed Diesel?"

- **Expected AI Answer:** Minimum 51.0 Cetane index for clean ignition.
- **Source Citation:** 07_quality_assurance_and_fuel_testing_standards.txt

Q28: "How does the Flue Gas Desulfurization (FGD) unit protect the environment?"

- **Expected AI Answer:** Removes 98.5% of SO2 exhaust emissions from Boilers B-201/B-202 to meet CPCB limits.
- **Source Citation:** 08_environmental_emissions_and_waste_treatment.txt

Q29: "What happens during a major quadrennial turnaround maintenance shutdown?"

- **Expected AI Answer:** Every 4 years, a 21-day full plant shutdown is executed for vessel internal inspection, tube retubing, and catalyst loading.
- **Source Citation:** 09_turnaround_maintenance_and_overhaul_manual.txt

Q30: "How is crude oil delivered and refined fuels dispatched from Northgate Refinery?"

- **Expected AI Answer:** Raw crude arrives via Marine Jetty-01 in 36-inch pipeline CF-101; refined fuels are dispatched via 60% pipeline, 48-wagon rail loading gantry, and 350 tank trucks per day.
- **Source Citation:** 10_supply_chain_and_crude_dispatch_logistics.txt